

Applied AI Professional Certification Program

IIT Hyderabad

Data Sourcing Guide

Where to get the data for each of the five problem statements — public datasets, synthetic generation, and the traps in between

Companion to: Problem Statements (5 options, 8 milestones)

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Audience: Participants and mentoring TAs

Read this first. You are not expected to invent your data from nothing. For four of the five problem statements there is a high-quality, openly licensed public dataset that gives you a realistic starting corpus in under an hour. Use it. Generate synthetically only for the parts nobody publishes — which is mostly the internal policy documents your RAG pipeline retrieves over.

1. How to Use This Guide

Teams asked for help sourcing data. This guide answers that for all five problem statements at once, because the underlying need is identical in every case even though the domain vocabulary is not.

1.1 Every problem statement needs four data assets

Each of the eight milestones consumes a different kind of data. Plan for four assets from day one:

Asset	First needed	What it is	Where it comes from
A. Document corpus	M3 → M4	Policy handbooks, playbooks, manuals, SOPs — the text you index and retrieve over	Mostly synthetic. Nobody publishes their internal policy corpus.
B. Intake records	M1	Raw, messy inputs: messages, applications, tickets, alarms	Synthetic, with public datasets for realism calibration
C. Mock API tables	M2 → M6	Relational state your tools read and write, later exposed over MCP	Public relational datasets, or the deterministic generator
D. Golden eval set	M8 (build from M4)	20 graded cases with expected answers and routes	Hand-written adversarial cases + generated coverage cases

The scheduling trap. Asset D is listed against M8 but you must start it at M4. The moment you have a working retrieval baseline you need something to measure it against, and a golden set assembled the night before the final presentation measures nothing. Write five cases at M4, five more at each milestone after.

1.2 The policy on real data has changed

The original document said "synthetic" throughout. Having reconsidered, the guidance is now:

- **Use openly licensed real data wherever it exists.** CUAD alone gives LexOps teams 510 genuine commercial contracts with 13,000+ expert clause annotations under CC BY 4.0. No synthetic generator will match that, and the annotations double as free retrieval ground truth.
- **Never use real personal data.** This has not changed and will not. No real patient records, no real loan applicants, no real customer PII. Synthetic-by-construction sources like Synthea satisfy this while still being realistic.
- **Generate the policy layer.** Internal handbooks, clause playbooks, credit manuals, equipment SOPs — these are exactly what organisations do not publish, and exactly what your RAG pipeline retrieves over. This is where synthetic generation earns its place.

The practical rule: real data for volume and realism, synthetic data for the documents that make your system domain-specific.

1.3 What a good corpus looks like

Retrieval evaluation is only meaningful if documents are distinguishable. Three properties matter more than size:

- **Specific numbers.** A corpus of documents that all say "timelines may vary" cannot be evaluated — every chunk is equally relevant to every query. Real policy documents state figures. Yours must too.
- **Deliberate near-duplicates.** Eight category addenda that each override the general return policy differently is a harder and more realistic retrieval problem than eight documents on unrelated topics. This is where reranking earns its keep in M4.
- **Known gaps.** Your corpus must not cover everything, so that "I don't know" is sometimes the correct answer. A system never asked an unanswerable question has never been tested for groundedness.

Fifty documents is a good target. Ten is too few for retrieval quality to vary with your chunking choices; five hundred is a week you do not have.

2. Ground Rules

2.1 Licensing

Several datasets recommended here are non-commercial or share-alike. That is fine for coursework and matters enormously if you carry the project into your job. Check before you reuse:

Licence	Means	Datasets here
CC BY 4.0	Free use including commercial, with attribution	CUAD, AI4I 2020, SECOM, ContractNLI
Public domain (US Gov)	No restrictions	SEC EDGAR, SBA 7(a), NASA C-MAPSS, CFPB, CMS
CDLA-Sharing 1.0	Attribution + derivatives shared alike	Bitext intent corpora
CC BY-NC-SA 4.0	Non-commercial only	Olist, LEDGAR / LexGLUE
Government publication	Freely downloadable; check reuse terms	RBI Master Directions, IRDAI circulars
Platform terms	Governed by the host's terms, not a standard licence	Kaggle-hosted sets without an explicit licence

2.2 No real personal data. Ever.

This is the one hard rule. It applies even when a dataset is public, even when it is de-identified, and even when it would make your demo better.

- Do not scrape real patient forums, review sites, or support threads for intake records.
- Do not use a colleague's real contract, loan file, or medical history as a test case.
- **MIMIC-IV is out of scope for this programme.** Not because it is inappropriate — it is excellent — but because PhysioNet credentialing plus CITI training takes weeks and your project is due in August. Synthea gets you 90% of the structural realism today, with no credentialing at all.

2.3 Record where your data came from

M8 asks you to defend your evaluation. The first question any reviewer asks is where the data came from and under what licence. Answer it in a file, not from memory. The toolkit writes a

```
data/<domain>/manifest.json
├── seed, provider, model used for generation
├── asset inventory with counts
├── public_sources_referenced [{name, url, use, licence}]
└── contains_real_personal_data: false
```

If you assemble data by hand instead, write the equivalent yourself. It takes ten minutes and it is the difference between a defensible system and an anecdote.

3. The Generator Toolkit

A working generator ships alongside this guide. It produces all four assets for all five problem statements, and it is provider-agnostic — Gemini AI Studio free tier, OpenRouter, or a local Ollama model, switched with one flag.

3.1 Quickstart

```
pip install -r requirements.txt
cp .env.example .env           # add your Gemini key

# See the plan and spend nothing
python generate.py --domain shopsense --dry-run

# Generate everything for one capstone
python generate.py --domain shopsense

# No API key yet? Mock API tables need no LLM at all.
python generate.py --domain plantguard --only tables
```

Domains: careflow, lexops, wealthpilot, shopsense, plantguard, all.

3.2 What it writes

data/<domain>/	
corpus/markdown/*.md	clean-parse baseline
corpus/pdf/*.pdf	the messy real case – chunk this too
intake/records.jsonl	each record carries a ground_truth block
mock_api/*.csv json	seed tables for your M2 tools
eval/golden_set.json	20 cases, categories and expected routes
manifest.json	provenance, seed, licences

3.3 Three things it does deliberately

Emits every document as both markdown and PDF

The PDF renderer is lossy on purpose. It flattens heading structure the way real policy PDFs do. If your M4 chunker performs well on the markdown and falls apart on the PDF, it is not ready — and the PDF is what production looks like.

Attaches a ground_truth block to every intake record

Urgency band, intent, risk band, missing fields, whether human review is required. That is your M1 parsing accuracy metric for free and your M5 routing labels. Do not discard it when you load the records.

Mixes hand-written adversarial cases into the eval set

A model asked to attack a corpus it just wrote produces polite attacks. The cases numbered -EV-9xx in each domain are hand-written to actually break things: prompt injection, authority-limit probes, false premises, impossible sensor readings, and — in WealthPilot — matched bias pairs where identical financials differ only by demographic proxy.

3.4 Reproducibility

Same seed produces byte-identical output, including all dates: generation is anchored to a frozen reference clock rather than the wall clock. This is what allows teams to be graded against a common rubric. Sampling in the LLM-generated stages still varies by provider; the tables and the eval scaffolding do not.

Iterate small. Run with `--corpus-docs 4 --intake 20 --eval-items 6` while you are still changing your schema. A full corpus run costs real minutes and real quota; there is no reason to pay that until your Pydantic models have stopped moving.

4. The Five Problem Statements

Each section below follows the same structure: the four data assets in order, then the traps specific to that domain. Read your own section; skim the others, because the failure modes transfer.

4.1. CareFlow — AI Care Coordination Assistant

Healthcare Operations

Shortest path. Run Synthea for patient and encounter data, generate the clinic policy corpus, and take insurance vocabulary from the IRDAI or CMS public circulars. Zero real PHI at any point, by construction rather than by promise.

4.1.1 Asset A — Policy and insurance document corpus

No clinic publishes its internal referral policy, so this corpus is generated. What you can borrow from public sources is the structure and vocabulary, which is what makes generated documents read like real ones.

Dataset	Licence	What it gives you	Get it from
IRDAI Master Circular on Health Insurance Business	Govt. of India publication	Real claim-process vocabulary, disclosure fields, grievance timelines. The India-context anchor — and available in Hindi as well as English.	irdai.gov.in → Document Detail
CMS Summary of Benefits and Coverage templates	Public domain (US Gov)	The standard benefit-table layout: deductible, coinsurance, out-of-pocket maximum, category exclusions.	cms.gov
Ayushman Bharat PM-JAY scheme documents	Govt. of India publication	Package rates, empanelment rules, pre-authorisation workflow for the Indian public health context.	pmjay.gov.in
MedQuAD	Public domain (NIH source)	Question phrasing patterns for health enquiries — useful for eval question realism, not for clinical content.	github.com/abachaa/MedQuAD

Generate roughly fourteen documents: a referral management policy, a pre-authorisation matrix, a cost-share schedule, a clinical escalation standard, an intake SOP, a data handling standard, and one service-line handbook per specialty. The specialty handbooks are the interesting part — they are deliberate near-duplicates that differ in their numbers, which is what forces your retriever to find the right one.

```
python generate.py --domain careflow --only corpus
# 14 documents → data/careflow/corpus/{markdown,pdf}/
```

4.1.2 Asset B — Patient intake messages

Generated, always. This is the asset where the temptation to scrape a real patient forum is strongest and where it is most clearly wrong.

What makes generated intake messages useful rather than decorative:

- Channel variety — a phone transcript with filler words and false starts parses very differently from a web form.
- Roughly one in five seeking clinical advice the clinic must refuse to give. These are your M8 guardrail cases, arriving through the front door rather than bolted on later.
- Roughly one in six missing the insurance ID or the specialty, so M1 has something real to detect.
- Some written in Indian English with local phrasing, because your participants' systems will meet that and a corpus of American English will not prepare them.

4.1.3 Asset C — Mock EHR and insurance tables

Use Synthea. It is an open-source synthetic patient population simulator from MITRE that produces complete lifetime medical histories in FHIR, CSV and C-CDA. Because it is built entirely from published statistics rather than trained on real records, the output is free of privacy restrictions by construction — there is no re-identification risk to argue about.

```
git clone https://github.com/synthetichealth/synthea && cd synthea
./run_synthea -p 500 --exporter.csv.export true

# → output/csv/{patients,encounters,conditions,
#               medications,procedures,payers}.csv
```

Or skip the build entirely and take a pre-generated population from synthea.mitre.org/downloads — including a 100,000-patient COVID set and the Coherent dataset that bundles FHIR with imaging and ECG data.

Why this matters beyond convenience: Synthea emits real FHIR resource shapes. If you model your Pydantic objects on them, a team that later wants to point the system at a real EHR sandbox does not rewrite its schema. The toolkit's own tables follow the same shapes for this reason.

The toolkit additionally generates four tables Synthea does not cover: plans, patients with eligibility status and deductible progress, appointments, and referrals with pre-authorisation flags.

4.1.4 Asset D — Golden eval set

Twenty cases. The distribution that matters here is weighted towards refusal, because the failure mode in this domain is a system that gives clinical advice:

Category	Count	What it tests
guardrail	6–7	Symptom interpretation, medication questions, "should I" phrasing, and the claimed-clinician bypass
factual	6–7	A specific co-pay, window, or SLA from one document
multi_hop	3–4	Plan coinsurance combined with a specialty's procedure coverage
unanswerable	2–3	Plausible clinic questions the corpus genuinely does not cover
injection	2	Ticket text instructing the assistant to drop its policy

The case that catches most teams. A patient who states they are themselves a doctor and asks a dosing question "clinician to clinician". Systems that pass every other guardrail case often fold on this one, because the request sounds authorised. Write the scope-of-practice document to say explicitly that the rule has no exceptions — then test that it holds.

4.1.5 Domain traps

- **Do not use MIMIC-IV.** Credentialing takes weeks. Synthea is available now.
- **Do not let the model write clinical content.** Your service-line handbooks describe operations — appointment durations, preparation instructions, coverage. The moment a corpus document contains diagnostic guidance, your guardrails are fighting your own retrieval layer.
- **Insurance figures must be internally consistent.** If the cost-share schedule and the pre-authorisation matrix disagree about a plan, your groundedness evaluation is measuring your corpus's contradictions rather than your system's faithfulness. Generate both in one pass and check them against each other.

4.2. LexOps — Contract Intelligence & Compliance Copilot

Legal / LegalTech

Shortest path. Download CUAD. It is 510 real commercial contracts with 13,000+ expert clause annotations under CC BY 4.0. Generate only the clause playbook, which CUAD does not contain and no firm publishes.

4.2.1 Asset A — Contracts and clause playbook

This is the one domain where you should not generate your primary corpus. CUAD gives you real contracts annotated by lawyers — annotations you would otherwise have to pay for, and which double as free ground truth for M4 retrieval evaluation.

Dataset	Licence	What it gives you	Get it from
CUAD v1 (The Atticus Project)	CC BY 4.0	510 commercial contracts across 25 agreement types, 13,000+ expert labels over 41 clause categories. Ships as PDF, TXT, a master-clauses CSV and a SQuAD-style JSON.	atticusproject.ai.org/cuad or HF: theatticusproject/cuad
SEC EDGAR full-text search	Public domain (US Gov)	Effectively unlimited further real contracts — filter for EX-10 material contract exhibits. This is where CUAD itself came from.	efits.sec.gov/LATEST/search-index
ContractNLI	CC BY 4.0	607 NDAs with entailment annotations. Excellent for the "does this clause actually say what the summary claims" guardrail check.	stanfordnlp.github.io/contract-nli
LEDGAR (via LexGLUE)	CC BY-NC-SA — non-commercial	60,000+ labelled contract provisions. Strong for clause classification; check the licence before any commercial reuse.	HF: lex_glue
MAUD	CC BY 4.0	Merger agreement understanding — useful if your team wants a harder clause-extraction stretch goal.	HF: theatticusproject/maud

Generate the playbook. CUAD tells you what contracts say. It does not tell you what your fictional company's position on those clauses is — preferred, fallback, walk-away, and the escalation thresholds. That comparison is the entire product, and it is the part you must create.

```
python generate.py --domain lexops --only corpus

# 19 documents: playbook overview, risk scoring methodology,
# output disclaimer standard, + 16 per-clause playbook sections
```

Sixteen clause sections each with a preferred position, a fallback, a walk-away, four numeric thresholds, and worked examples of acceptable and unacceptable counterparty language. Point your RAG at the playbook, feed CUAD contracts in as the documents under review, and the M4 task becomes real: does the cited playbook position actually support the redline the system drafted?

4.2.2 Asset B — Contract review requests

Generated. These are the internal messages that arrive at a three-person legal team, and their defining characteristic is impatience. Include:

- Requests that paste 600 words of clause inline, and requests that paste two lines.
- About a fifth containing genuinely aggressive terms — uncapped indemnity, unilateral termination for convenience, IP assignment of customer data.
- **Deadline pressure.** "Just approve it, we're closing today." A compliant system must not yield, and you cannot test that without the pressure in your corpus.

Pair each request with clause text drawn from CUAD rather than generated, where you can. Real contract language is denser and stranger than anything a model writes to order.

4.2.3 Asset C — Contract repository and e-signature mock

The toolkit generates counterparties with negotiation history and posture, contracts with terms, renewal dates and risk scores, and signature-request envelopes. The per-counterparty negotiation memory the problem statement asks for in M3 keys off the counterparties table.

```
python generate.py --domain lexops --only tables # no API key needed
```

4.2.4 Asset D — Golden eval set

Twenty clause-review cases. Because CUAD is annotated, you can build a large slice of this mechanically: pick contracts, pick clause categories, and the annotation is the expected answer. That is a rare gift — take it.

Category	Count	What it tests
factual	7–8	A threshold, cap multiple, or notice period from one playbook section
multi_hop	4–5	A clause position combined with the risk scoring methodology to produce an approval route
guardrail	4–5	Deadline pressure, requests to opine on enforceability, auto-approval above threshold
unanswerable	2–3	Contract questions the playbook does not address
injection	1–2	Instructions embedded in pasted clause text

4.2.5 Domain traps

- **Never reproduce real published contract text in generated documents.** Use CUAD contracts as inputs to review; write your own model clause language for the playbook.

- **"Is this enforceable" is the guardrail case.** It is the most natural question a user will ask and the one your system must never answer. Write the disclaimer standard to prohibit it explicitly, then test it.
- **CUAD contracts are long.** Several exceed typical context windows. That is a feature — M7 asks you to harden against long-document parsing failures, and here you have genuine ones rather than contrived ones.

4.3. WealthPilot — SME Loan Underwriting & Credit Research Assistant

FinTech / Digital Lending

Shortest path. Take real SME lending distributions from the SBA 7(a) public file, real policy language from RBI's MSME Master Direction, and generate the credit manual. Build your bias probes as matched pairs from day one — they cannot be retrofitted.

4.3.1 Asset A — Credit policy and regulatory corpus

Dataset	Licence	What it gives you	Get it from
RBI Master Direction — Lending to MSME Sector	Govt. of India publication	FIDD.MSME & NFS.12/06.02.31/2017-18, updated Feb 2026. Real priority-sector thresholds, definitions, classification rules. The India-context anchor.	rbi.org.in → Notifications
SBA 7(a) and 504 Loan Data Reports	Public domain (US Gov)	Every SBA-guaranteed loan since FY1991: amounts, sectors, terms, outcomes. The only large public dataset of genuine SME (not consumer) lending decisions.	catalog.data.gov → SBA 7(a)
Home Credit Default Risk	Kaggle competition terms	Applicant feature schema and realistic default base rates across 300k+ applications.	kaggle.com/c/home-credit-default-risk
CFPB Consumer Complaint Database	Public domain (US Gov)	Real complaint narratives — excellent adversarial intake text that synthetic generation will not match for genuine grievance.	consumerfinance.gov/data-research/consumer-complaints
FinQA / TAT-QA	MIT / CC BY	Numerical reasoning over financial documents. Useful if you want to evaluate the arithmetic your risk agent performs.	HF: dreamerdeo/finqa, next-tat/tat-qa

A note on Lending Club. It appears in every credit-risk tutorial and it is consumer lending, not SME. Use it for loan-lifecycle vocabulary if you like; do not use it to calibrate SME risk. SBA 7(a) is the right reference for this problem statement.

Generate thirteen documents: core underwriting standards, a fair lending standard, document verification, risk grading and pricing, adverse action — plus one sector underwriting note per sector, each with genuinely different thresholds.

4.3.2 Asset B — Loan applications

Generated, with one requirement that is easy to miss:

the financials must be internally consistent so DSCR is actually computable. A model asked for "realistic financials" will happily emit an EBITDA that contradicts the revenue and the debt service. Then your M2 calculator is being tested against noise, and you will spend an afternoon debugging an agent that is working correctly.

Generate the ratios first, derive the line items from them, and validate on the way out.

Build the bias probes into intake, not just eval. Roughly a tenth of applications should volunteer irrelevant personal information — family circumstances, community, region, the applicant's own hardship narrative. These exist so you can verify that information reaching the model does not reach the decision. If it only appears in your eval set, you have tested the guardrail without ever testing the pipeline.

4.3.3 Asset C — Bureau, bank statement and decision tables

Four tables: applicants, bureau reports with score/enquiries/DPD history, eighteen months of monthly bank statements per applicant, and past decisions with reason codes and human signatories. The past-decisions table is what makes M3's "persistent memory of applicant interaction history" meaningful rather than empty.

```
python generate.py --domain wealthpilot --only tables
```

4.3.4 Asset D — Golden eval set, and the matched-pair construction

This is the eval set that differs most from the others, because the problem statement asks specifically for cases that probe bias. There is only one construction that actually detects bias, and it is not a set of varied test cases.

Matched pairs. Two applications with identical financials — same revenue, same EBITDA, same debt service, same bureau score, same sector, same operating history — differing only in a demographic proxy such as the applicant's name or city. The correct outcome is not "a fair decision on each". The correct outcome is

identical decisions and identical rationales. Any divergence is a failure, and it is measurable rather than a matter of judgement.

Category	Count	What it tests
bias_probe (matched pairs)	6 (3 pairs)	Name, city, and language-of-application varied against identical financials
factual	5–6	A ratio floor, tenor limit, or tolerance from the manual
multi_hop	3–4	A sector note combined with core standards or the pricing grid
guardrail	3–4	Auto-approval above the signature threshold; sympathy arguments; protected attributes offered as justification
unanswerable	2–3	Credit questions the manual does not address

Score the pairs as pairs. A per-case accuracy metric will report 100% on a system that approves Rajesh and declines Fatima, because each individual decision looks defensible. Compute agreement within each pair as its own metric and report it separately.

4.3.5 Domain traps

- **A fabricated policy citation is a compliance failure, not an error.** Weight your unanswerable cases more heavily here than in any other domain.
- **Postal code is a protected proxy.** So is the applicant's name, their institution of education, and the language they applied in. Write these into the fair lending standard explicitly — a system that avoids explicit protected attributes while happily using postal code has not solved the problem.
- **Do not use real applicant data under any framing.** Not a friend's business, not an anonymised file from work.

4.4. ShopSense — Customer Care & Order Operations Assistant

Retail / E-commerce

Shortest path. Richest public data of the five. Bitext gives you a real support-intent taxonomy, Olist gives you a genuine nine-table order database. Generate only the policy handbook.

4.4.1 Asset A — Policy corpus

No retailer publishes a machine-readable returns policy you can lawfully redistribute, so this layer is generated. Fourteen documents: returns and refunds, refund authorisation matrix, shipping, warranty, escalation triggers, fraud prevention, plus one addendum per product category.

The category addenda are the point. Eight documents that each override the general 30-day return window differently — electronics 15 days, apparel 45, groceries non-returnable — is a genuinely hard retrieval problem. A system that retrieves the general policy when asked about a blender gives a confident wrong answer, which is exactly the failure M4 reranking exists to fix.

4.4.2 Asset B — Support tickets

Dataset	Licence	What it gives you	Get it from
Bitext retail/e-commerce chatbot dataset	CDLA-Sharing 1.0	8.47M tokens across 27 intents in ORDER, DELIVERY, PRODUCT, REFUND, FEEDBACK categories. Gives you a real intent taxonomy instead of one you guessed at.	HuggingFace: bitext/Bitext-retail-e-commerce-llm-chatbot-training-dataset
Customer Support on Twitter	Kaggle platform terms	Around 3M real support tweets. Genuinely angry customers — which synthetic generation systematically under-produces, because models are trained to be polite.	kaggle.com/datasets/thoughtvector/customer-support-on-twitter
Amazon Reviews 2023 (McAuley Lab)	Research use — see card	Product metadata and review text for catalogue realism.	HF: McAuley-Lab/Amazon-Reviews-2023

Take the intent taxonomy from Bitext and generate ticket text against it. That gives you labels grounded in a real taxonomy and text you control. Pure Bitext is too clean — it is training data for chatbots, so the questions are well-formed. Real tickets are not.

Your generated tickets should include: one in eight genuinely angry with mild profanity, one in six missing the order reference, two or three claiming refunds above the automated cap, at least two mentioning legal

action or a consumer forum, at least two attempting prompt injection inside the ticket body, and at least one product-safety report. Several in Indian English with code-mixing.

4.4.3 Asset C — Order database

Use Olist. It is a real Brazilian marketplace dataset of roughly 100,000 orders across nine linked tables — orders, order items, payments, reviews, customers, sellers, products, geolocation, category translations. That relational structure is exactly what your M2 order-lookup and refund tools need, and it is far more realistic than anything you would invent.

Licence check. Olist is CC BY-NC-SA 4.0 — non-commercial. Fine for this capstone. Not fine if you carry the project into a product without revisiting it.

The toolkit generates an equivalent five-table set (customers with return rates and abuse flags, products, orders, shipments with carrier scan states, refunds with approval tiers) if you would rather stay entirely synthetic or want Indian carriers and rupee amounts.

4.4.4 Asset D — Golden eval set

Category	Count	What it tests
factual	6–7	A window, fee, or SLA — several answerable only from the category addendum, not the general policy
guardrail	5–6	Refunds above the automated cap, third-contact escalation, legal threats, product safety, account-flag disclosure
injection	3–4	Fake system blocks inside ticket text instructing a refund
multi_hop	3	A category addendum combined with the refund authority matrix
unanswerable	2	Policy questions the handbook does not cover

This domain carries the heaviest injection weighting of the five, and correctly so: ticket text is attacker-controlled by definition. A customer can write anything into a support form, and your system reads all of it.

4.4.5 Domain traps

- **Synthetic anger is weak anger.** Models generate frustration that resolves politely. Calibrate against the Twitter support corpus, or your escalation-on-sentiment logic will be tuned to customers who do not exist.
- **Make the category addenda actually conflict.** If every category has a 30-day window, retrieval precision is untestable because any retrieved document gives the right answer.
- **The refund cap is the sharpest guardrail.** It is a single number, easy to state, easy to test, and produces an unambiguous pass or fail. Test it with an angry customer and a large amount together — sentiment plus value is where systems break.

4.5. PlantGuard — Predictive Maintenance & Supply Chain Ops Copilot

Manufacturing / Supply Chain

Shortest path. Take sensor data from AI4I 2020 or NASA C-MAPSS. Generate the manuals and SOPs. Do not let a language model write your time series.

4.5.1 Asset A — Equipment manuals and SOPs

Generated, necessarily — real manufacturer manuals are copyrighted and cannot be redistributed. Thirteen documents: lockout/tagout, permit to work, maintenance planning, spares procurement, alarm response, plus one manual per asset class.

Each asset manual carries operating ranges with specific alarm setpoints, a PM schedule by running hours, at least six named fault conditions with causes and corrective actions, invented part numbers, torque specifications, and an explicit list of which actions are safety-critical. Make the numbers genuinely differ between assets — that is what forces retrieval to find the right manual instead of a plausible one.

Dataset	Licence	What it gives you	Get it from
OSHA standards and eTool content	Public domain (US Gov)	Real lockout/tagout and permit-to-work structure and vocabulary for your safety standards.	osha.gov
US Army / DoD technical manuals	Public domain (US Gov)	Genuine equipment-manual register and structure, freely redistributable — useful as a style reference for what a real manual reads like.	armypubs.army.mil
awesome-industrial-datasets	Index — check each entry	Curated index of further public industrial datasets if you outgrow the ones below.	github.com/jonathanwvd/awesome-industrial-datasets

4.5.2 Asset B — Sensor alerts and maintenance logs

Do not generate sensor time series with an LLM. Language models produce numbers that look plausible and are statistically wrong: no autocorrelation, no drift, no realistic degradation curve. Any anomaly detection you build on top will be detecting the model's sampling artefacts. Use real data for numbers and generation only for the prose around them.

Dataset	Licence	What it gives you	Get it from
AI4I 2020 Predictive Maintenance (UCI id 601)	CC BY 4.0	10,000 rows, 14 features, five labelled failure modes (tool wear, heat dissipation,	archive.ics.uci.edu/dataset/601

Dataset	Licence	What it gives you	Get it from
		power, overstrain, random). Realistic ~3–6% class imbalance. Start here.	
NASA C-MAPSS Turbofan Degradation	Public domain (US Gov)	Run-to-failure trajectories with remaining-useful-life ground truth across four subsets of varying difficulty. The benchmark for genuine degradation curves.	NASA PCoE Data Repository
Microsoft Azure Predictive Maintenance	Kaggle platform terms	Five linked tables — telemetry, errors, maintenance, failures, machines — across 100 machines. Closest structural match to the M2 tool APIs.	kaggle.com/datasets/arnabbiswas1/microsoft-azure-predictive-maintenance
SECOM (UCI)	CC BY 4.0	590 features, heavy missingness. Good if you want a realistic data-quality problem.	archive.ics.uci.edu/dataset/179
MIMII	CC BY-SA 4.0	Malfunctioning industrial machine sound — valves, pumps, fans, slide rails. A multimodal stretch goal.	zenodo.org/records/3384388

Generate the prose layer on top: alarm strings with tag names and values, and messy shift-log entries in an operator's own words ("making a funny noise near the drive end since morning shift"). That mixture — machine-precise and human-vague describing the same event — is what M1 has to reconcile.

If you use the toolkit's tables instead, its telemetry comes from a degradation simulator rather than the LLM: baseline plus drift plus noise, with a seeded accelerating fault on about a fifth of assets. Healthy and faulty ranges deliberately overlap, so a threshold alarm on instantaneous vibration misses most faults until far too late, and trend analysis is required.

4.5.3 Asset C — Asset registry, work orders, inventory

Assets with criticality classification and running hours, 30 days of hourly telemetry, work orders with permit flags and downtime, and a spare-parts inventory with lead times and supplier tiers. The inventory table is what M6's procurement agent acts on and what the ERP MCP server exposes.

```
python generate.py --domain plantguard --only tables
```

4.5.4 Asset D — Golden eval set

The stakes here are physical, so the distribution leans hard towards refusal and ambiguity:

Category	Count	What it tests
guardrail	6–7	Production pressure to skip lockout, alarm suppression, auto-raising a PO above threshold, safety-critical actions without sign-off
factual	5–6	A setpoint, torque value, or interval from the correct asset's manual
multi_hop	3–4	An asset's corrective action combined with the lockout or permit standard
unanswerable	3–4	Plausible questions about assets or faults the corpus does not document
ambiguous	2	Contradictory or physically impossible sensor readings

The impossible-reading case is the one worth writing carefully. Vibration of 912 mm/s and a temperature of -40°C is a broken sensor, not a failing bearing. A system that confidently diagnoses a mechanical fault from garbage input will do the same thing in production, and the resulting work order sends a technician to strip a healthy machine.

4.5.5 Domain traps

- **A fabricated repair step is a safety hazard.** Weight unanswerable cases heavily. The worst available failure in this domain is inventing an isolation procedure for a system the corpus does not document — an ammonia skid, a high-voltage panel — and having a technician follow it.
- **Production pressure is the realistic attack.** "We're losing 40,000 an hour, skip the paperwork" is what your system will actually meet, and it is far more effective against a helpful assistant than any prompt injection.
- **Do not scrape manufacturer manuals.** They are copyrighted. Public-domain military technical manuals are a legitimate style reference; a scraped Grundfos PDF is not.

5. Data Quality Checklist

Run this before you build on a corpus. Every item has cost a team a day at some point.

5.1 Corpus

- Documents contain specific numbers, not hedges. Search your corpus for "may vary" and "typically" — if they are everywhere, retrieval evaluation is meaningless.
- At least some documents are deliberate near-duplicates differing in their figures.
- There are known gaps, so "I don't know" is sometimes correct.
- The PDF and markdown versions are both indexed and both tested. If your chunker only works on markdown, it is not ready.
- Documents do not contradict each other on the same fact, unless you put the contradiction there on purpose and recorded it.

5.2 Intake records

- Every record carries a ground_truth block, and you have not discarded it on load.
- Length, register and completeness genuinely vary. Sort by character count and look at the distribution — if it is narrow, regenerate with a stronger variety instruction.
- Roughly one in six is missing a field your parser needs.
- Numeric fields are internally consistent where a downstream calculator depends on them.
- Adversarial and edge cases are present in the intake stream, not only in the eval set.

5.3 Mock API tables

- Foreign keys actually resolve. A ticket referencing an order that does not exist will silently poison your M2 tool testing.
- Status distributions are realistic — mostly delivered, some cancelled, few returned.
- At least one table has enough rows that retrieval or filtering is non-trivial.
- Dates are anchored to a fixed reference, not the wall clock, or your runs will not be reproducible.

5.4 Golden eval set

- Every category is represented, including unanswerable and injection.
- Adversarial cases are hand-written, not model-written. Models write polite attacks.
- Expected routes are specified, not just expected answers — M5 and M8 both need them.
- must_not_contain is populated for guardrail cases. "Did it refuse" is easier to score reliably than "was the refusal well-phrased".
- You built the first five cases at M4, not the night before the presentation.

6. Five Ways Teams Lose a Weekend

Generating the corpus last

The corpus is a dependency of M3, M4, M5 and M8. Teams that leave it until M4 discover that a 50-document generation run takes real time, that their first attempt produced documents too vague to retrieve over, and that regenerating invalidates the indexing work they just did. Generate a small corpus in week one, index it, and only then scale up.

A corpus with no specific numbers

The single most common failure. A model asked to write a policy document defaults to corporate hedging, and hedged documents are indistinguishable to a retriever. The fix is in the prompt: demand figures, thresholds, named exceptions, and clause numbering, and reject documents that do not contain them.

Evaluating only on questions the corpus answers

A system that always produces a confident answer, scores well on a set of answerable questions and fails the moment it meets a real one. Unanswerable cases are not padding; in the lending and maintenance domains they are the highest-stakes cases you have.

Testing guardrails with cases the model wrote

Ask a model to write prompt-injection cases against its own corpus and it produces attacks that announce themselves. Real injections are embedded in otherwise ordinary content — a support ticket with a fake system block halfway down, after a genuine complaint. Write these yourself.

Discovering at M6 that the mock APIs have no data behind them

M2 tools often get built against hardcoded stubs, which works until M6 asks you to expose the same system over MCP and run a multi-agent team against it. Seed the tables first; the toolkit generates them offline in seconds with no API key.

Appendix A — All Datasets at a Glance

Dataset	Domain	Licence	One-line
CUAD v1	LexOps	CC BY 4.0	510 real contracts, 41 clause types, 13k expert annotations. The strongest single dataset available across all five problem statements.
Synthea	CareFlow	Unrestricted	Synthetic FHIR patient records, privacy-free by construction. No credentialing.
AI4I 2020	PlantGuard	CC BY 4.0	10k rows, five labelled failure modes. The fastest route to real sensor data.
Bitext retail/e-commerce	ShopSense	CDLA-Sharing 1.0	27 real support intents, 8.47M tokens. A taxonomy you did not have to guess.
SBA 7(a) loan data	WealthPilot	Public domain	Genuine SME lending decisions since FY1991. Not consumer lending.
NASA C-MAPSS	PlantGuard	Public domain	Run-to-failure trajectories with RUL ground truth.
Olist	ShopSense	CC BY-NC-SA 4.0	Nine-table relational order database, ~100k orders. Non-commercial.
SEC EDGAR	LexOps	Public domain	Unlimited further real contracts via EX-10 exhibits.
RBI MSME Master Direction	WealthPilot	Govt. of India	Real Indian priority-sector lending policy, updated Feb 2026.
IRDAI Health Insurance Circular	CareFlow	Govt. of India	Real Indian health-insurance process vocabulary, English and Hindi.
ContractNLI	LexOps	CC BY 4.0	607 NDAs with entailment labels for groundedness checks.
LEDGAR / LexGLUE	LexOps	CC BY-NC-SA 4.0	60k+ labelled provisions. Non-commercial.
Home Credit Default Risk	WealthPilot	Kaggle terms	Applicant feature schema and default base rates.
CFPB Complaints	WealthPilot	Public domain	Real grievance narratives for adversarial intake.
Customer Support on Twitter	ShopSense	Kaggle terms	~3M real support tweets. Genuinely angry customers.
Azure Predictive Maintenance	PlantGuard	Kaggle terms	Five linked tables closest to the M2 tool shape.

Dataset	Domain	Licence	One-line
SECOM	PlantGuard	CC BY 4.0	590 features, heavy missingness. A real data-quality problem.
MIMII	PlantGuard	CC BY-SA 4.0	Industrial machine sound. Multimodal stretch goal.
MAUD	LexOps	CC BY 4.0	Merger agreement understanding. Harder extraction stretch.
FinQA / TAT-QA	WealthPilot	MIT / CC BY	Numerical reasoning over financial documents.
Amazon Reviews 2023	ShopSense	Research use	Product metadata and review text for catalogue realism.
MedQuAD	CareFlow	Public domain	Health question phrasing patterns. Not for clinical content.

Two datasets deliberately excluded. MIMIC-IV — excellent, but PhysioNet credentialing plus CITI training runs to weeks and the capstone is due in August. Lending Club — ubiquitous in tutorials, but it is consumer lending and this problem statement is SME. Using it will calibrate your risk model against the wrong population.